

Answer all questions.

1. (a) (i) State Ohm's law.

[1]

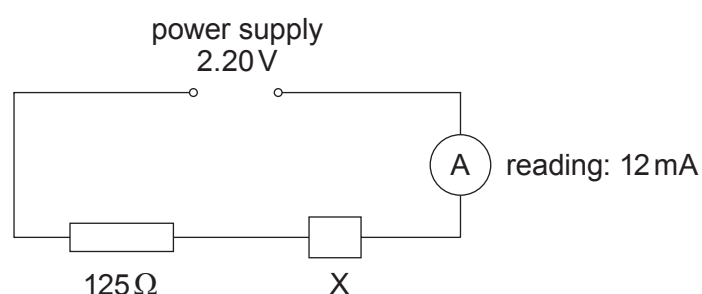
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- (ii) What can be said about the *resistance* of a conductor that obeys Ohm's law? [1]

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- (b) (i) An electrical component, X, is included in the circuit shown. The internal resistance of the power supply is negligible.



Show that the resistance of X in this circuit is approximately $60\ \Omega$.

[2]

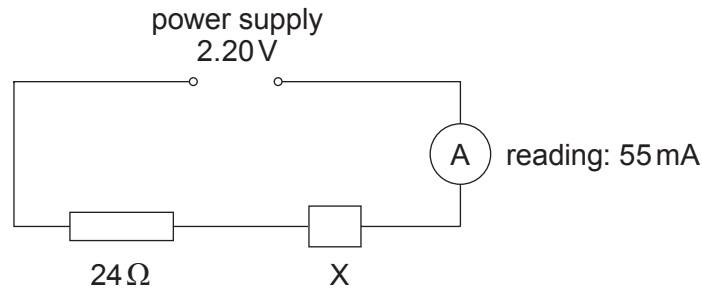
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- (ii) When the 125Ω resistor in (b)(i) is replaced by a 24Ω resistor, the reading on the ammeter increases, as shown below.



Evaluate whether or not X obeys Ohm's law, presenting your argument clearly. [2]

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- (iii) State, giving a reason, whether or not X could be a filament lamp. [1]

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- (c) A certain high temperature superconductor has a transition temperature of -188°C . The boiling point of liquid nitrogen is -196°C .

- (i) State what is meant by the *transition temperature* of a superconductor. [1]

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- (ii) Give one possible use for a high temperature superconductor and state why it would be an advantage for the transition temperature to be above the boiling point of liquid nitrogen. [2]

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